**DO NOW — RECALL**

# Lesson 101 · Multicore CPUs and Parallel Systems

Name: _____ Class: _____ Date: _____

**1. What is a core?**

**2. Define concurrency.**

**3. What does Amdahl's law illustrate?**

**4. A 3×4 array of independent cells: how many cells?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A CPU processing unit able to run a hardware thread or more, depending on design.

### 2. Model answer

Several tasks progress over overlapping time.

### 3. Model answer

Serial work limits parallel speedup.

### 4. Model answer

12.

### By the end, you can…

• A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

• Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

• Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

### Vocabulary — know these terms

multicore — Processor with multiple CPU cores.

parallelism — Simultaneous processing of independent tasks.

critical path — Longest dependent chain.

Amdahl's law — Speedup bounded by serial portion.

NEW LEARNING

# Read, annotate and explain

### Understand

A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

### Apply

Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

### Evaluate

Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Eight equal independent jobs are shared across two identical cores. How many ideal waves?**

### Explain the choice

8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Twelve equal independent jobs run on three identical cores. Compare ideal waves with one core. [6]

### Question 2

Explain two benefits and two limitations of multicore processing for video rendering. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Half a program must run serially; other half parallelised perfectly across four cores. Compute ideal speedup and interpret. [6]

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Four CPU cores always give exactly four times the performance of one core.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.